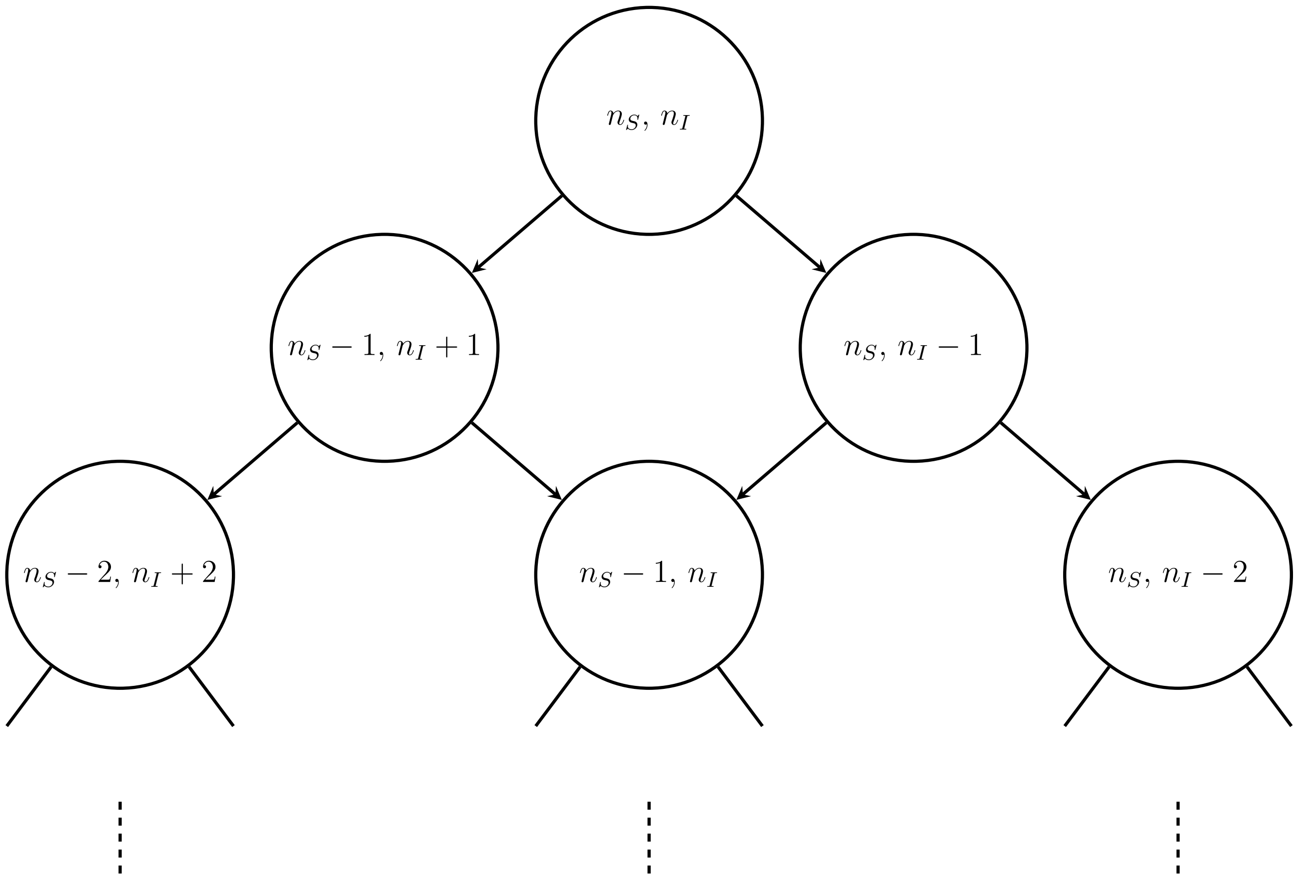
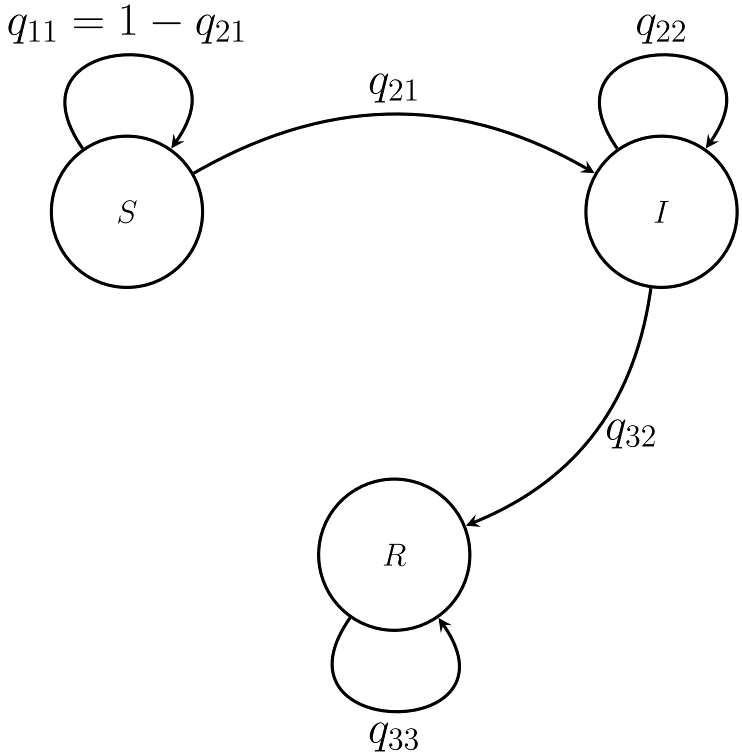
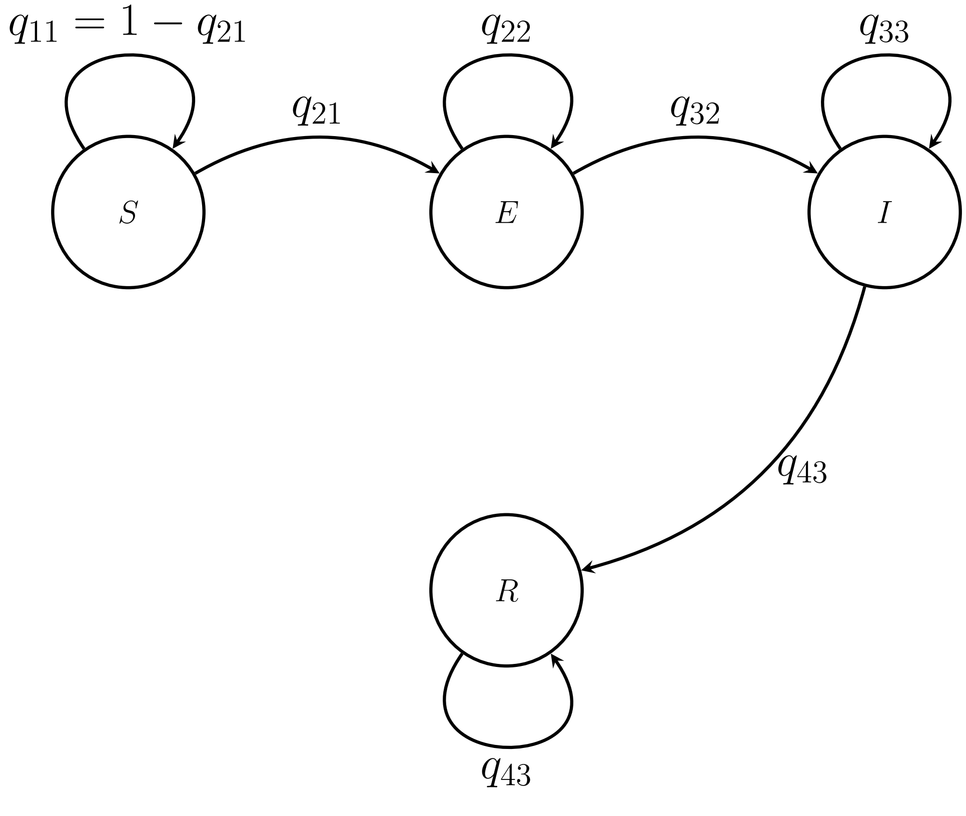
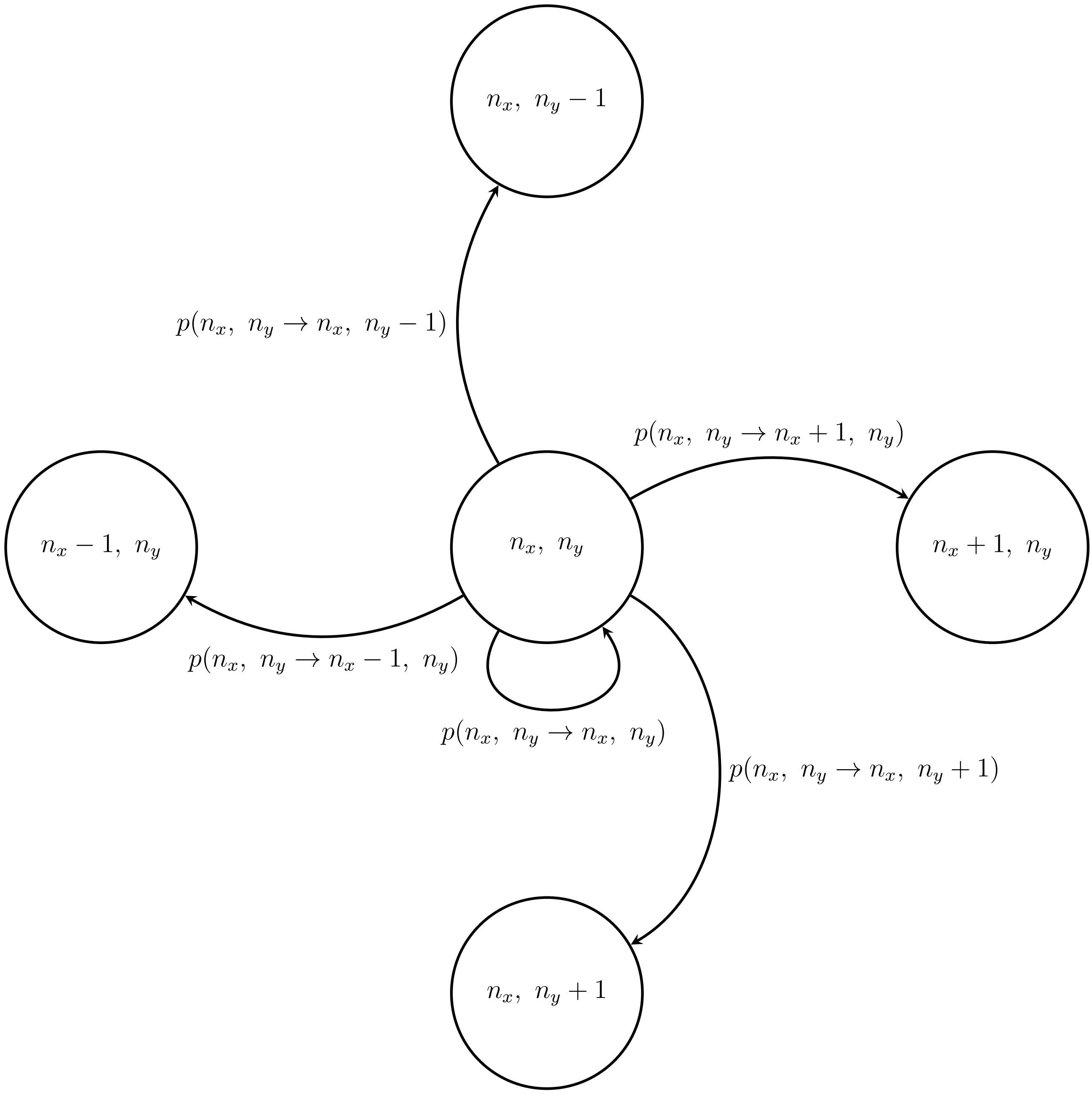
1 Markov Chain Diagrams




$q_{11} = 1 - q_{21}$
q_{22}
q_{21}
S
I
q_{32}
R
q_{33}


$q_{11} = 1 - q_{21}$
q_{22}
q_{33}
q_{21}
q_{32}
S
E
I
q_{43}
R
q_{43}



$n_x,\ n_y - 1$
$p(n_x,\ n_y \to n_x,\ n_y - 1)$
$p(n_x,\ n_y \to n_x + 1,\ n_y)$
$n_x - 1,\ n_y$
$n_x,\ n_y$
$n_x + 1,\ n_y$
$p(n_x,\ n_y \to n_x - 1,\ n_y)$
$p(n_x,\ n_y \to n_x,\ n_y)$
$p(n_x,\ n_y \to n_x,\ n_y + 1)$
$n_x,\ n_y + 1$

2 Plots

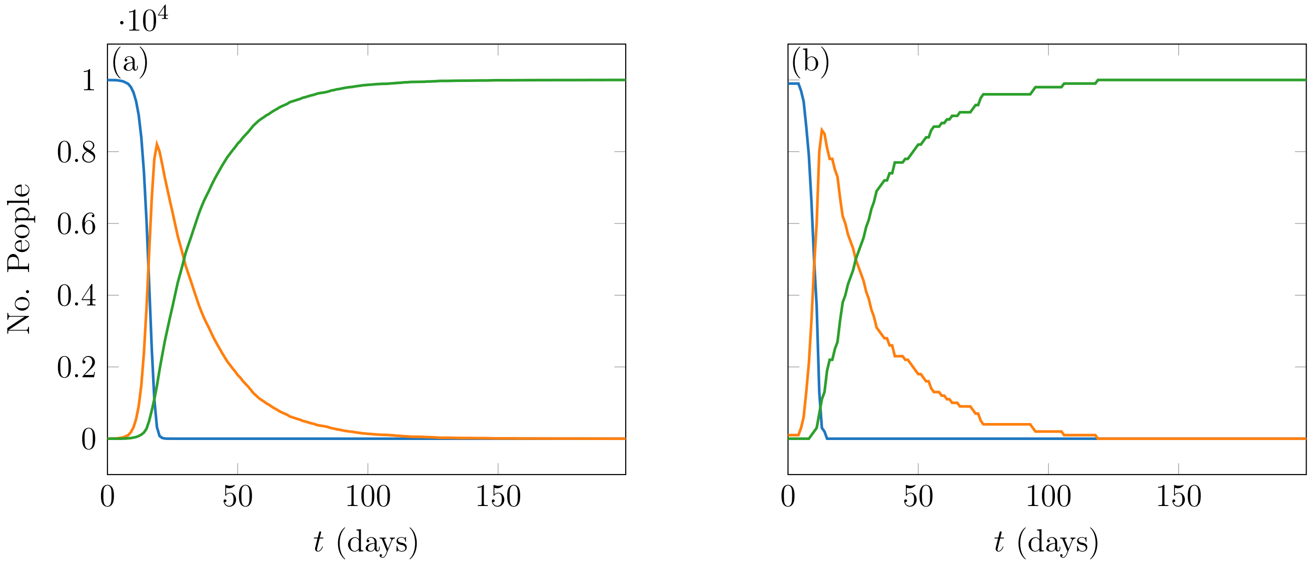


Figure 1: Microscopic SIR model with 100,000 people. System starts from 1 infection and 99,999 susceptible.

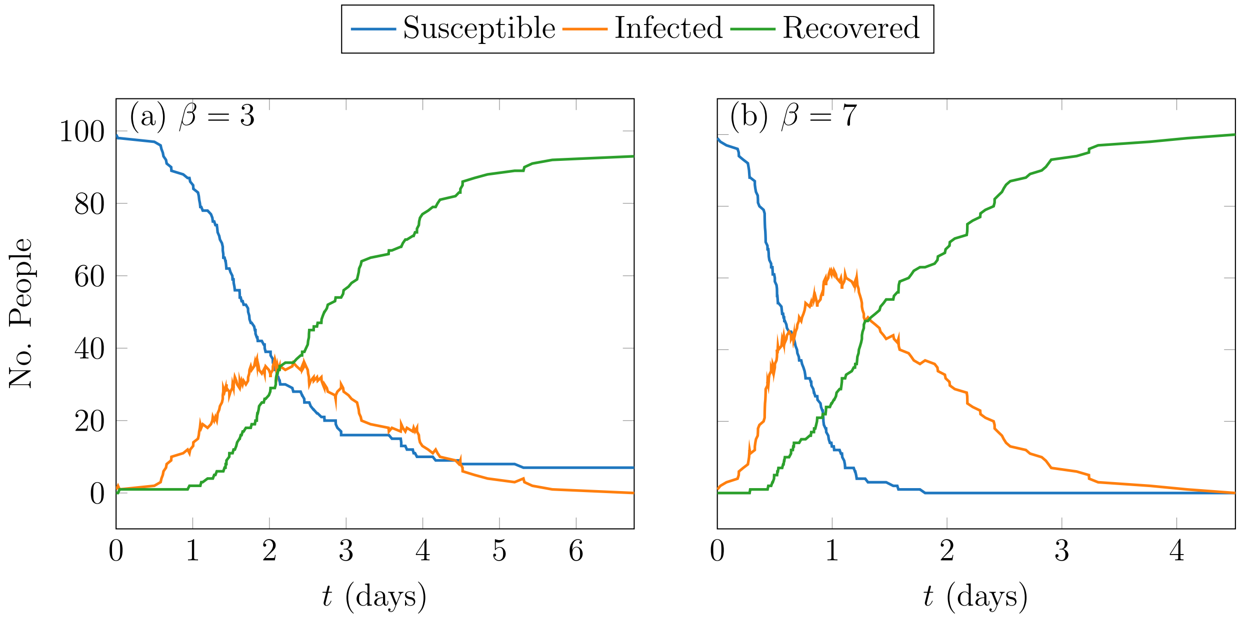


Figure 2: SIR model with β/N people infected per day per infected person per susceptible person and γ people recovered per day per infected person, where N is the total population. (a): $\beta = 3$, $\gamma = 1$; (b): $\beta = 7$, $\gamma = 1$. Figures produced by M. G. Hauge Evans.

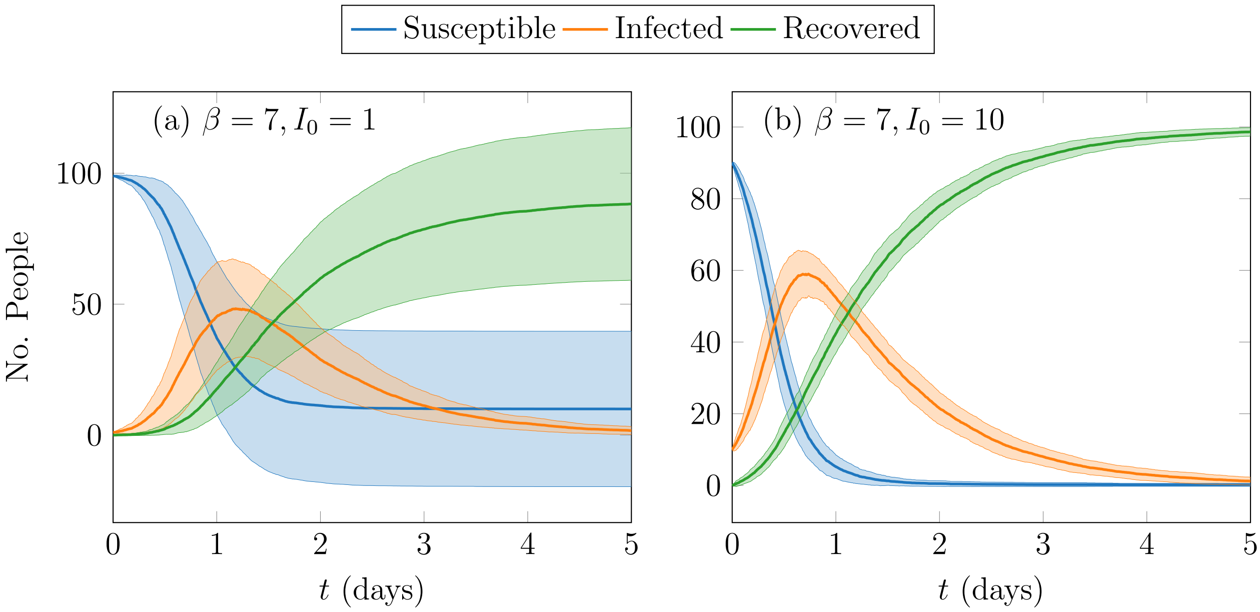


Figure 3: Ensemble averaged SIR model with β/N people infected per day per infected person per susceptible person and $\gamma = 1$ people recovered per day per infected person, where $N = 100$ is the total population. I_0 people are infected at $t = 0$ and the rest of the population is susceptible. (a): $\beta = 7$, $I_0 = 1$; (b): $\beta = 7$, $I_0 = 10$. Figures produced by M. G. Hauge Evans.

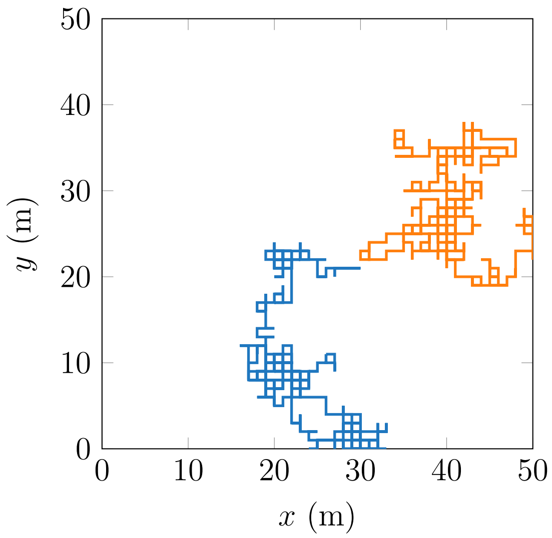


Figure 4: Random Walk Plot. Figures produced by M. G. Hauge Evans.